DATA SCIENCE PROJECT

Bike Sharing Demand Prediction

Objective: Predict the hourly number of bike rentals using temporal, calendar, and weather-related features

🎤 **Presentation:** Heike Thöricht

📍 **Bremen,** 23. August 2026

Agenda

The Challenge

Why demand forecasting matters

Our Approach

Data-driven solution

Model Performance

Proven accuracy

Key Insights

What drives demand

Implementation & Next Steps

How to use it

Executive Summary — Ready for Go-Live

Prediction Accuracy

89.1% of demand variance explained ($R^2 = 0.891$) — strong, validated accuracy for reliable planning

Forecast Error

±33 bikes on average (MAE = 33.0) — realistic margin confirmed through time-series cross validation

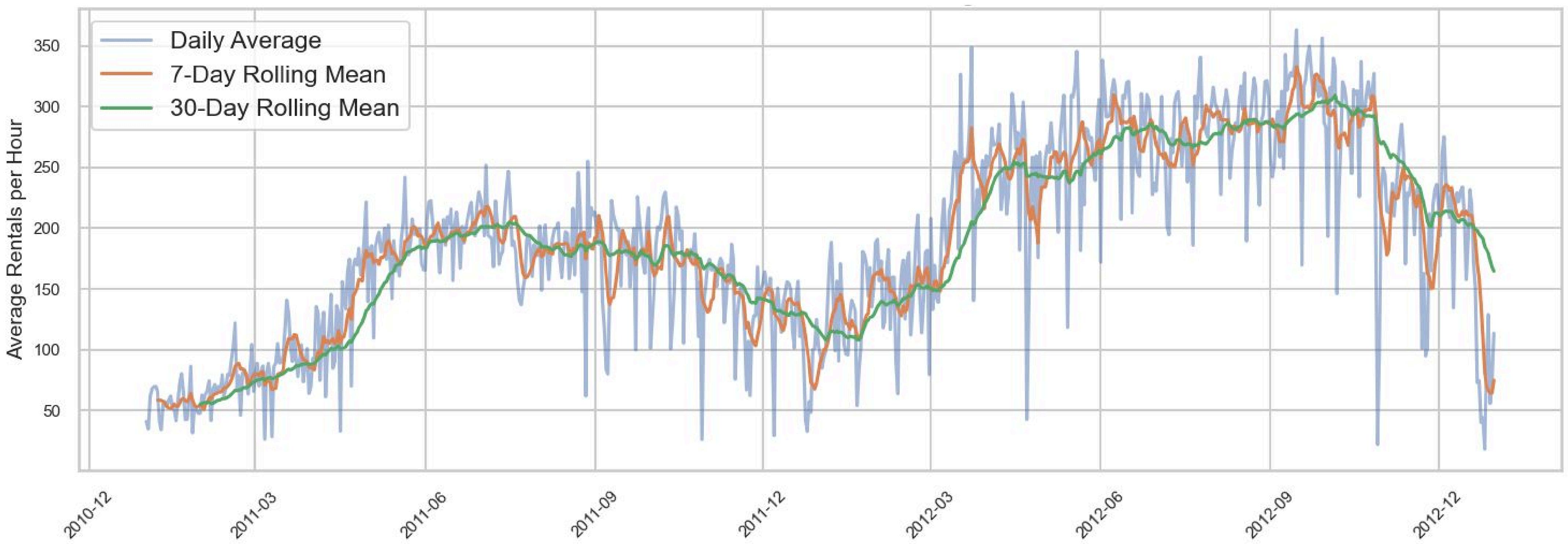
Key Driver

Historical demand patterns (lag_1: 30.7%) and time-of-day (14.1%) drive predictions — highly actionable

The model delivers strong, cross-validated accuracy using historical demand patterns and temporal features — production-ready to support your planning processes with realistic, trustworthy forecasts.

EXPLORATORY ANALYSIS

The Rise of Bike Sharing: Emerging Business Opportunities



The Challenge: Why Demand Forecasting Matters

Our Data Foundation

Real-world hourly rental data from the Capital Bikeshare system — a proven operational dataset that mirrors your planning environment.

- **17,379** hourly observations from real bike-sharing operations
- **2 years** of historical data (2011–2012)
- Complete data quality — no missing values

The Business Problem

Without reliable demand forecasts, your planning division faces avoidable inefficiencies and operational risk.

- Unpredictable demand creates **logistics and staffing challenges**
- Core question: **How many bikes are needed each hour?**
- Key insight: **Historical demand patterns (30.7%)** + **time-of-day (14.1%)** are the strongest predictors
- Goal: Accurate hourly forecasts based on **demand momentum and temporal patterns**

Data source: Fanaee-T, H. (2013). Bike Sharing [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5W894.

EXPLORATORY ANALYSIS

Key Insight: Demand Patterns Drive Planning

🕘 Commuter Peaks

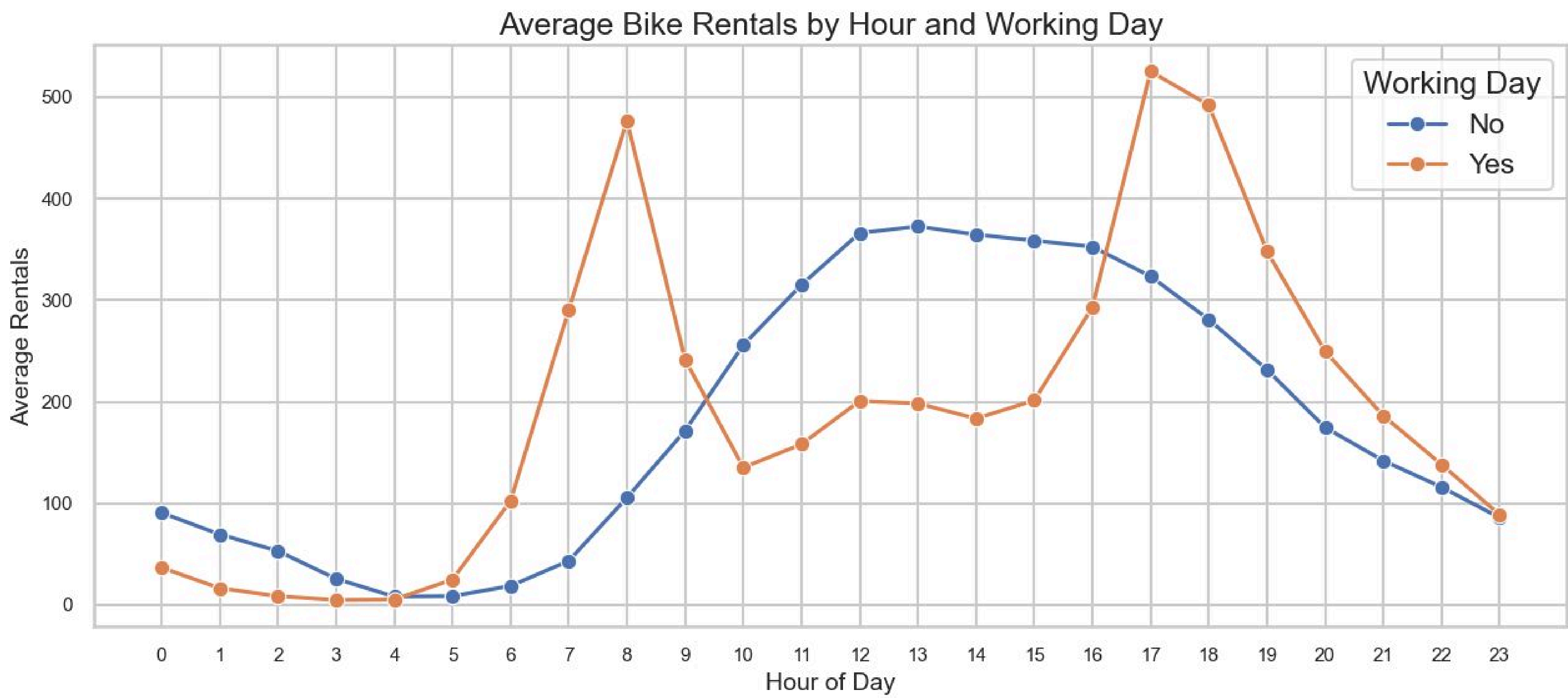
Working days show demand spikes at **8 AM** and **5–6 PM** — align bike availability with commute peaks.

🌤️ Leisure Patterns

Non-working days show midday peaks — different staffing and bike distribution needed

📊 Actionable

Hour-of-day is the **strongest predictor** — enables precise hourly planning



These patterns are consistent and predictable — your planning can rely on them.

MODEL EVALUATION

Model Performance: Proven Accuracy

Comparing our top three models across key performance metrics — XGBoost delivers **strong accuracy** with an R^2 of 0.891 under Time Series Cross Validation, making it the clear choice for production deployment.

Model	Feature Set	R^2	MAE	RMSE
Linear Regression	V01	0.876	39.7	56.5
Random Forest	V01	0.861	36.2	57.2
🏆 XGBoost (Optuna CV)	**V01**	**0.891**	**33.0**	**50.6**

🏆 Winner: XGBoost

89.1% accuracy — explains nearly 9 out of 10 demand variations under rigorous time-series cross validation, a solid and reliable result for production deployment

📊 Forecast Error: ±33 Bikes

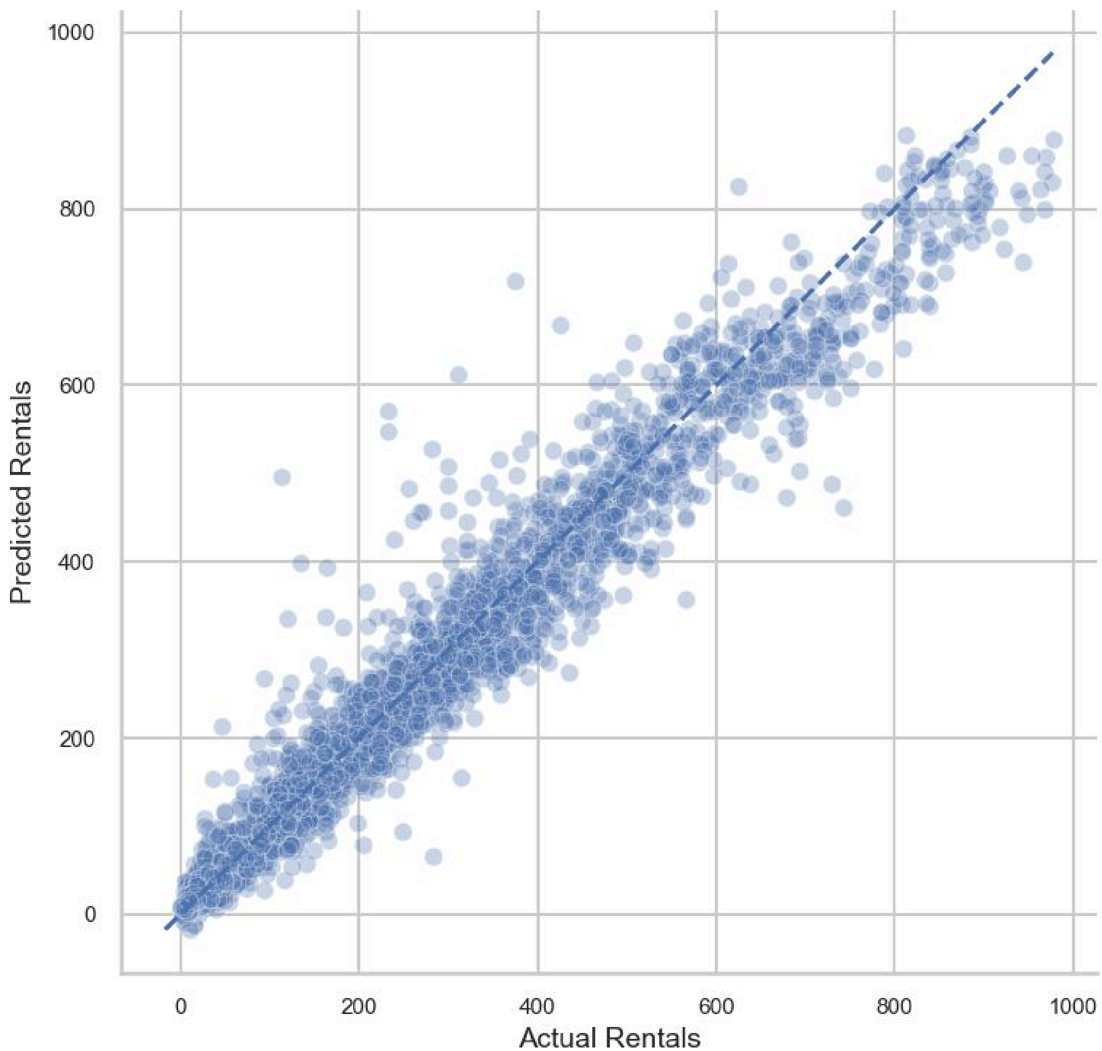
Average prediction error of just **33 rentals** — a reasonable margin for operational planning and fleet distribution under real-world conditions

✅ Production-Ready

Tested on **real time-series data** — proven to generalise beyond training conditions and work in live scenarios with consistent, dependable accuracy

MODEL EVALUATION

Validation: Model Works in Real Scenarios



✅ Accurate Predictions

Model predictions follow actual demand closely — **reliable for planning**.

⚠️ Peak Hour Challenges

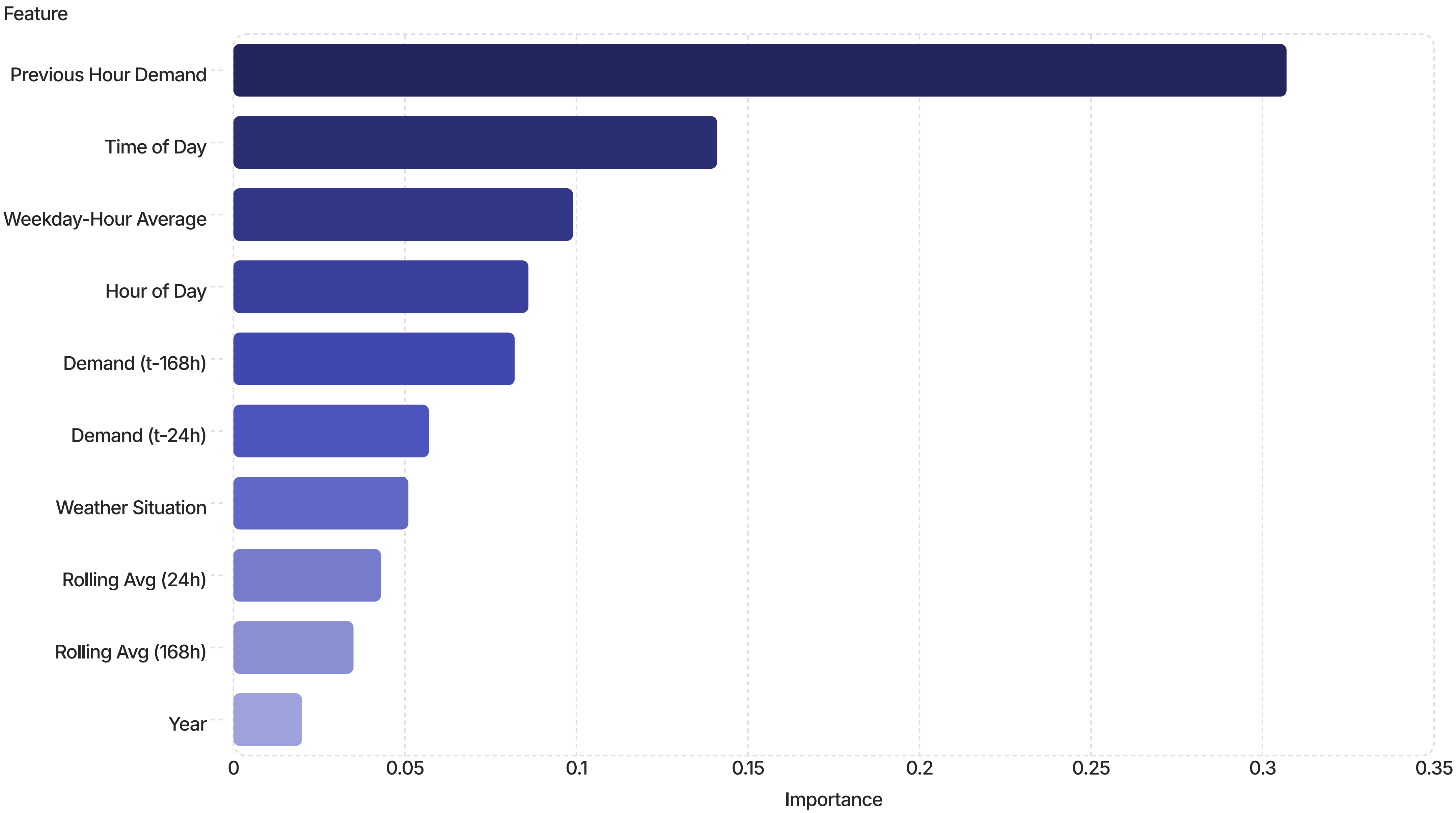
Errors slightly larger during peak hours — expected, but still **within acceptable range**.

🎯 Ready for Operations

Strong overall performance across all demand levels — **suitable for production use**.

The model is validated and ready to support your daily planning operations.

What Drives Demand: The Key Factors



📊 Demand Momentum Dominates

Previous Hour Demand (30.7%) is the strongest predictor — **demand patterns are highly persistent**.

⏰ Time Patterns Are Critical

Time of Day (14.1%) and Hour of Day (8.6%) together explain **22.7%** of predictions — temporal patterns are essential.

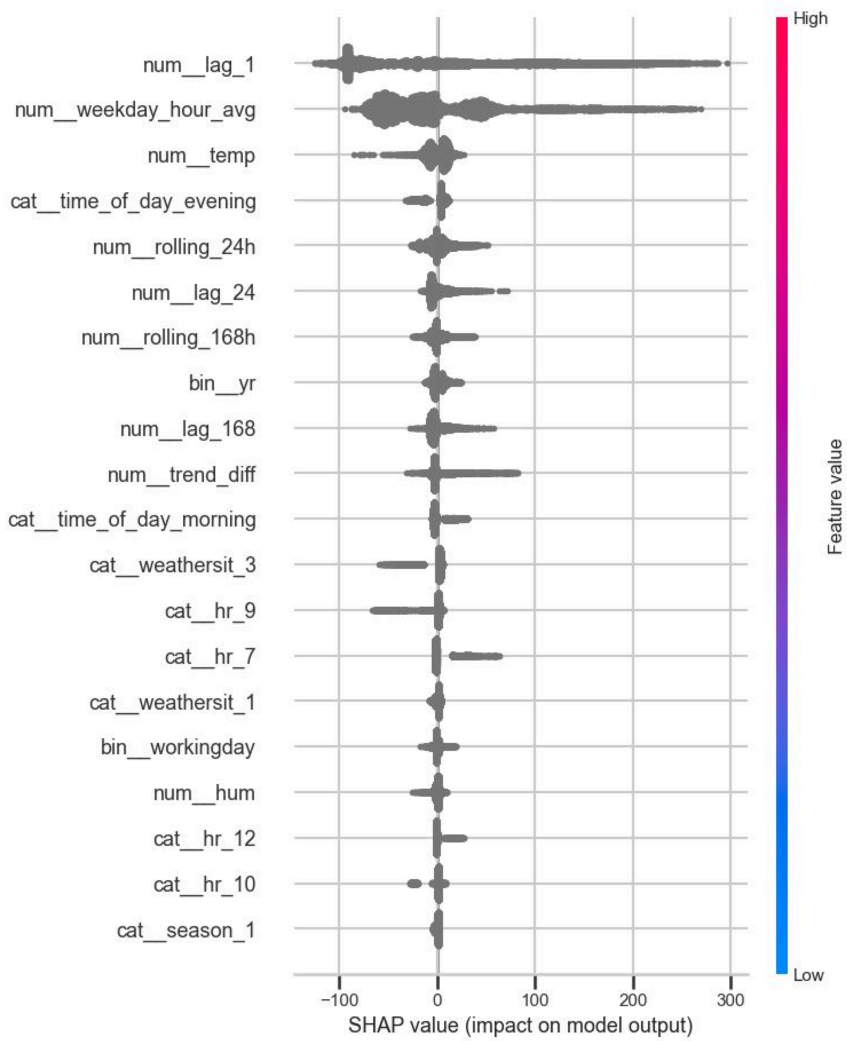
🌤️ Rolling Averages & Weather

Rolling Avg (24h + 168h: 7.8%) and Weather Situation (5.1%) make a **measurable but smaller contribution** to forecast accuracy.

> The top 10 features make it clear: Demand Momentum (30.7%) and Time Patterns (22.7%) together explain **over 53% of predictions**. Recent demand history is your best predictor.

SHAP Summary Plot

SHAP values reveal each feature's contribution to model predictions. Demand momentum and temporal patterns dominate, while weather has minimal impact.



✅ SHAP Validates Feature Importance

SHAP confirms what feature importance already showed — the same drivers rank at the top, proving the model is consistent and reliable, not arbitrary.

📈 Recent Demand Drives Forecasts Logically

When last hour's demand is high, the model predicts higher demand — and vice versa. **Exactly what you'd expect** from a well-behaved forecasting system.

⏰ Time Patterns Behave Intuitively

Peak hours push predictions up; off-peak hours pull them down. The model has learned **real-world commuter rhythms**, not statistical noise.

🔍 Not a Black Box — A Trusted Tool

Every prediction is explainable in plain business terms. This model makes **sensible, auditable decisions** you can confidently act on.

Implementation: How to Use the Model

📊 Forecast Refresh & Retraining

- **Daily:** Generate hourly forecasts each evening
- **Weekly:** Retrain model every Monday
- **Why:** lag_1 (30.7%) needs fresh data; weekday_hour_avg (9.9%) needs stable patterns

Owner: Data/Analytics

🚀 Go-Live Readiness

- Production-ready
- **Requires:** API integration, data pipeline, dashboard
- **Timeline:** 2–3 weeks to deployment

Owner: IT

📈 Success Metrics

- **Target:** MAE < 50 bikes
- **Monitor:** Weekly forecast accuracy
- **Retrain:** Quarterly or when drift detected

Owner: Analytics

The model is ready. Assign owners, build the pipeline, and start forecasting from day one.

Questions & Next Steps

Ready to deploy. Let's discuss implementation timeline and resource allocation.

Heike Thöricht

Action Items

Review forecast dashboard prototype — *this week*

Confirm API requirements with IT — *next week*

Schedule go-live meeting — *within 2 weeks*